Lab1

1-GitHub account:

https://github.com/AhmedAmer10

2-Download and install docker on your machine

Install Docker Engine on RHEL:

Step1: uninstall old versions

```
sudo dnf remove docker \
        docker-client \
        docker-client-latest \
        docker-common \
        docker-latest \
        docker-latest-logrotate \
        docker-logrotate \
        docker-engine \
        podman \
        runc
```

Step2: set up the repository

```
sudo dnf -y install dnf-plugins-core
sudo dnf config-manager --add-repo
https://download.docker.com/linux/rhel/docker-ce.repo
```

step3: install Docker Engine:

3.1: Install the Docker packages

sudo dnf install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin

3.2: Start Docker Engine

sudo systemctl enable --now docker

3.3: Verify that the installation is successful by running the hello-world image:

sudo docker run hello-world

```
[root@server ~]# sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
17eec7bbc9d7: Pull complete
ea52d2000f90: Download complete
Digest: sha256:ef54e839ef541993b4e87f25e752f7cf4238fa55f017957c2eb44077083d7a6a
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/

[root@server ~]#
[root@server ~]#
```

Q1:

1- Pull nginx:alpine image on your machine

docker pull nginx:alpine

```
[root@server ~]# docker pull nginx:alpine
alpine: Pulling from library/nginx
5e7756927bef: Pull complete
589002ba0eae: Pull complete
bca5d04786e1: Pull complete
3e2c181db1b0: Pull complete
6b7b6c7061b7: Pull complete
955a8478f9ac: Pull complete
399d0898a94e: Pull complete
6d397a54a185: Pull complete
dda176b4ea00: Download complete
8a29e1f4877b: Download complete
Digest: sha256:1d13701a5f9f3fb01aaa88cef2344d65b6b5bf6b7d9fa4cf0dca557a8d7702ba
Status: Downloaded newer image for nginx:alpine
docker.io/library/nginx:alpine
[root@server ~]#
```

2- run nginx:alpine image on your machine in the background and specify the name of the container to name-ITI -< NO>

docker run -d --name Ahmed-ITI-64 nginx:alpine

```
[root@server ~]# docker run -d --name Ahmed-ITI-64 nginx:alpine
8a7e4ee2092631e66c7f8a48e44d2e6f2ebe157f8de563eed4929e0da5459323
[root@server ~]#
```

3- Run apache container with apache image with memory limit 70 MB and 1 core of cpu

docker run -d --name apache-container --memory="70m" --cpus="1.0" httpd:alpine

```
[root@server ~]# docker run -d --name apache-container --memory="70m" --cpus="1.0" httpd:alpine
Unable to find image 'httpd:alpine' locally
alpine: Pulling from library/httpd
abd702cd48c8: Pull complete
79273eb2147b: Pull complete
4f4fb700ef54: Pull complete
b9df760bab4a: Pull complete
987f24f891f4: Pull complete
3688fd767e4a: Pull complete
2bd78f43169f: Download complete
8c8ca383ec3c: Download complete
Digest: sha256:8f26f33a7002658050e9ab2cd6b77502619dfc89d0a6ba2e9e4a202e0ef04596
Status: Downloaded newer image for httpd:alpine
58bc7530155aa36fc4164c0c0088e91730b0a450d9bb91cf21e765cef3a72a05
[root@server ~]#
```

4- make sure to get the path of the cgroups created for this container and memory limit file and value

- **find the ID of the container**

  docker ps

  docker inspect --format '{{.Id}} {{.Name}}' apache-container

```
[root@server ~]#
[root@server ~]# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED         STATUS         PORTS     NAMES
58bc7530155a   httpd:alpine   "httpd-foreground"       3 minutes ago   Up 3 minutes   80/tcp    apache-container
8a7e4ee20926   nginx:alpine   "/docker-entrypoint.…"   6 minutes ago   Up 6 minutes   80/tcp    Ahmed-ITI-64
[root@server ~]#
[root@server ~]# docker inspect --format '{{.Id}} {{.Name}}' apache-container
58bc7530155aa36fc4164c0c0088e91730b0a450d9bb91cf21e765cef3a72a05 /apache-container
```

- **the path of the cgroups (v2) created for this container**

  /sys/fs/cgroup/system.slice/docker-<container-id>.scope/

- **Read memory limit value**

  cat /sys/fs/cgroup/system.slice/docker-58bc7530155aa36fc4164c0c0088e91730b0a450d9bb91cf21e765cef3a72a05.scope/memory.max

```
[root@server ~]#
[root@server ~]# cat /sys/fs/cgroup/system.slice/docker-58bc7530155aa36fc4164c0c0088e91730b0a450d9bb91cf21e765cef3a72a05.scope/memory.max
73400320
```

Memory=73400320=70M

Q2: Run an alpine container called memory-eater with memory limit of 50M, make sure that the container does not use swap memory as well construct the container such that it should consume more than 50M of memory, observe and prove that container is being killed by the OOM killer

Command:

```
docker run --name memory-eater \
  --memory="50m" \
  --memory-swap="50m" \
  alpine \
  sh -c "cat /dev/zero | head -c 104857600 | tail"
```

- **Memory = swap-memory=50M** (to make sure that the container does not use swap memory)
- **cat /dev/zero | head -c 104857600** | tail (to make the container attempted to consume 100MB which is more than the memory size=50MB.

Result: Container output is **Killed**

```
[root@server ~]#
[root@server ~]# docker run --name memory-eater \
  --memory="50m" \
  --memory-swap="50m" \
  alpine \
  sh -c "cat /dev/zero | head -c 104857600 | tail"
Killed
[root@server ~]# docker inspect memory-eater --format='{{.State.OOMKilled}}'
true
[root@server ~]#
[root@server ~]#
```

Verification: docker inspect memory-eater --format='{{.State.OOMKilled}}'

⇨ true